AIRLINES FLIGHT DATA ANALYSIS

A Mini Project Report

Submitted in partial fulfillment of requirement of the

Degree of

BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING

BY

Khyati Jain - EN23CS301637

Muskan Panjabi - EN23CS301642

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Under the Guidance of

Prof. Sonal Modh Bhardwaj

Prof. Rashmi Choudhary

Department of Computer Science & Engineering

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MEDI-CAPS UNIVERSITY, INDORE- 453331

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Report Approval

The project work "**AIRLINES FLIGHT DATA ANALYSIS**" is hereby approved as a creditable study of an engineering/computer application subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.

It is to be understood that by this approval the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion drawn therein; but approve the "Project Report" only for the purpose for which it has been submitted.

Internal

Examiner Name:

Prof. Sonal

Modh Bhardwaj

Prof. Rashmi

Choudhary

Designation

Affiliation

External

Examiner Name:

Designation

Affiliation

Declaration

We hereby declare that the project entitled **"Airlines Flight Data Analysis"** submitted in partial fulfillment for the award of the degree of Bachelor of Technology/Master of Computer Applications in 'B.Tech CSE' completed under the supervision of **Prof. Sonal Modh Mentor, Prof. Rashmi Choudhary,** Faculty of Engineering, Medi-Caps University Indore is an authentic work.

Further, we declare that the content of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for the award of any degree or diploma.

Khyati Jain - EN23CS301637
Muskan Panjabi - EN23CS301642
Kritika Geel - EN23CS301638

Signature and name of the student(s) with date

Certificate

We, **Prof. Sonal Modh Mentor, Prof. Rashmi Choudhary** certify that the project entitled **"Airlines Flight Data Analysis"** submitted in partial fulfillment for the award of the degree of Bachelor of Technology/Master of Computer Applications by **Khyati Jain, Kritika Geel, Muskan Panjabi** is the record carried out by them under my/our guidance and that the work has not formed the basis of award of any other degree elsewhere.

Prof. Rashmi Choudhary,

Prof. Sonal Modh Bhardwaj,

Computer Science & Engineering

Medi-Caps University, Indore

<Name of External Guide (If any)>

<Name of the Department>

Name of the Organization

Dr. Kailash Chandra Bandhu

Head of the Department

Computer Science & Engineering Medi-

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I would like to express my deepest gratitude to the Honorable Chancellor, **Shri R C Mittal,** who has provided me with every facility to successfully carry out this project, and my profound indebtedness to **Prof. (Dr.) D. K. Patnaik,** Vice Chancellor, Medicaps University, whose unfailing support and enthusiasm has always boosted up my morale. I also thank **Dr.Ratnesh Litoriya,** Dean, Faculty of Engineering, Medicaps University, for giving me a chance to work on this project. I would also like to thank my Head of the Department **Dr.Kailash Chandra Bandhu** for his continuous encouragement for the betterment of the project.

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Without their support this report would not have been possible.

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Abstract

This project focuses on the analysis of flight data to extract meaningful insights and patterns from large datasets. The main objective of this project is to understand flight trends, delays, cancellations, and overall performance of airlines. The dataset used in this project contains information such as flight timings, departure and arrival details, delays, and airline data.

Various data analysis techniques have been applied using tools like Python, Pandas, and data visualization libraries to clean, process, and analyze the data. The project involves identifying key factors that affect flight delays and performance.

The results obtained from the analysis help in understanding patterns in flight operations and provide useful insights for improving efficiency and reducing delays. This project demonstrates how data analysis can be effectively used in the aviation industry for better decision-making and performance optimization.

Keywords:

- Flight Data
- Data Analysis
- Airline Performance
- Flight Delays
- Data Visualization
- Python
- Pandas
- Aviation Industry
- numpy
- Matplotlib
- SeaBorn

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Abbreviations:

Abbreviation	Full Form
UI	User Interface
DB	Data Base
DFD	Data Flow Diagram
UML	Unified Modeling Language
CRUD	Create,Read,Update,Delete
IDE	Integrated Development Environment
HTML	HyperText Markup Language
CSS	Cascading Style Sheets

CHAPTER 1

INTRODUCTION

The aviation industry generates a large amount of data every day due to continuous flight operations across different airlines and airports. With the increasing number of flights, managing and analyzing this data has become an important task for improving efficiency, reducing delays, and enhancing passenger experience. Flight Data Analysis helps in studying this large amount of information to extract useful insights and patterns.

The main objective of this project is to analyze flight-related data and understand various factors affecting flight performance such as delays, cancellations, and scheduling issues. By using data analysis techniques, it becomes possible to identify the reasons behind delays and find patterns related to airlines, time, weather, and other operational factors.

In this project, a dataset containing flight information is used. This dataset includes details such as airline name, departure and arrival times, delay duration, flight routes, and cancellation status. The data is processed and analyzed using Python programming language along with powerful libraries such as Pandas, NumPy, Matplotlib, and Seaborn. These tools help in cleaning the data, performing calculations, and creating visual representations for better understanding.

Exploratory Data Analysis (EDA) is performed to study the dataset in detail. Different types of visualizations such as bar charts, line graphs, and histograms are used to represent trends and relationships within the data. These visualizations help in identifying important insights such as peak delay periods, performance comparison between airlines, and frequency of cancellations.

The scope of this project is to provide meaningful insights that can help airlines improve their operational efficiency. By understanding patterns in flight delays and performance, airlines can take better decisions to optimize schedules and reduce passenger inconvenience. This type of analysis is also useful for airport authorities and aviation planners.

Overall, this project demonstrates how data analysis techniques can be applied in the aviation sector to improve decision-making and performance. It highlights the importance of data-driven approaches in solving real-world problems and enhancing service quality in the airline industry.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Literature review is an essential part of any research or project work as it provides an understanding of the existing work carried out in the related field. It helps in identifying various approaches, techniques, and methodologies used by researchers and also highlights the research gaps.

In the field of aviation, large amounts of data are generated daily due to flight operations. This data includes information about flight schedules, ticket prices, delays, routes, and passenger traffic. Analyzing such data helps in improving airline performance, optimizing pricing strategies, and enhancing customer satisfaction.

In recent years, with the advancement of data science and analytics, flight data analysis has gained significant importance. Researchers and analysts are using various tools and techniques to extract meaningful insights from large datasets.

2.2 Overview of Flight Data Analysis

Flight data analysis involves studying historical flight data to understand patterns and trends in airline operations. The primary goal is to extract useful information that can help in decision-making.

Various aspects of flight data analysis include:

- Analysis of ticket pricing
- Study of flight delays
- Evaluation of airline performance
- Analysis of routes and traffic patterns

With the help of data analysis, airlines can improve their operational efficiency and provide better services to passengers.

2.3 Previous Work in Flight Data Analysis

Several studies have been conducted in the field of flight data analysis. Researchers have used different approaches to analyze flight data and solve real-world problems.

One of the major areas of research is **flight delay analysis**. Studies have shown that delays are influenced by factors such as weather conditions, airport congestion, and airline efficiency.

Another important area is **ticket price prediction**. Researchers have used statistical methods and machine learning algorithms to predict ticket prices based on historical data. Factors such as duration, number of stops, airline type, and demand play a significant role in determining ticket prices.

Many studies have also focused on **route analysis**, where data is analyzed to identify high-demand routes and optimize flight scheduling.

These studies highlight the importance of data analysis in the aviation industry and demonstrate how it can be used to improve decision-making.

2.4 Tools and Technologies Used in Literature

Various tools and technologies have been used by researchers for flight data analysis. Some of the commonly used tools include:

- **Python:** Widely used for data analysis due to its simplicity and powerful libraries

Among these, Python is the most popular choice because of its extensive library support. Libraries such as Pandas and NumPy are used for data manipulation, while Matplotlib and Seaborn are used for visualization.

These tools help in transforming raw data into meaningful insights through graphs, charts, and statistical analysis.

2.5 Factors Affecting Flight Ticket Prices

Based on the literature, several factors influence flight ticket pricing. Some of the important factors are:

- **Airline Type:** Premium airlines generally charge higher prices
- **Duration:** Longer flights may have higher prices
- **Number of Stops:** Direct flights are more expensive than connecting flights
- **Demand:** High demand leads to higher prices
- **Booking Time:** Prices increase as the departure date approaches

Understanding these factors is important for analyzing and interpreting flight data.

2.6 Comparative Analysis of Existing Approaches

Different approaches have been used in previous studies to analyze flight data. These include:

- Statistical analysis
- Data visualization techniques
- Machine learning models

Statistical analysis provides basic insights, while visualization techniques help in understanding patterns more clearly. Machine learning models are used for prediction purposes.

However, visualization-based analysis is often considered the first step before applying complex models, as it helps in understanding the dataset.

2.7 Research Gap

Although significant research has been carried out in the field of flight data analysis, most of the existing work focuses on prediction using machine learning techniques.

However, there is a lack of simple and clear exploratory data analysis (EDA) that helps in understanding the dataset before applying advanced techniques.

This project focuses on performing detailed exploratory data analysis using visualization techniques to identify patterns and trends in flight data.

2.8 Conclusion of Literature Review

From the literature review, it is evident that data analysis plays a crucial role in understanding airline operations and improving efficiency. Various techniques have been used by researchers to analyze flight data, including statistical methods, visualization, and machine learning.

This project builds upon the existing work by focusing on exploratory data analysis using Python. The insights obtained from this analysis can be useful for both passengers and airlines.

Chapter 3:METHODOLOGY

Introduction 3.1

Purpose The purpose of this Software Design Specification (SDS) is to describe the design and functional structure of the Airline Flight Data Analysis system. This document provides a clear description of the software components, system architecture, and data processing methods used to analyze airline flight data. It serves as a reference for understanding how the system processes flight-related datasets to generate meaningful insights and analytical results. The scope of this system focuses on analyzing airline flight information such as flight schedules, delays, cancellations, departure and arrival times, and other operational parameters. The system is designed to process large datasets and perform data cleaning, transformation, and visualization to identify patterns and trends in airline operations. This SDS specifically covers the design and implementation of the data analysis module, including data preprocessing, statistical analysis, and visualization components. The document outlines how the system transforms raw flight data into structured information that can support better understanding of airline performance and operational efficiency.

3.2 Document Conventions

This document follows a structured format to ensure clarity, consistency, and easy understanding of the software requirements for the Airline Flight Data Analysis system. Standard documentation practices have been followed while preparing this Software Requirements Specification (SRS). Headings and subheadings are used to organize the document into clearly defined sections. Major sections are written in bold to highlight their importance, while subsections provide detailed explanations of specific aspects of the system. Requirement statements are written in clear and precise language to avoid ambiguity.

Each requirement is described independently so that it can be easily understood, implemented, and verified. Where necessary, numbered lists and bullet points are used to present information in a structured manner. Technical terms related to data analysis, software components, and system functionality are used consistently throughout the document. Any assumptions or dependencies related to the system are explicitly mentioned within their respective sections. In this document, higher-level requirements define the overall functionality of the system, while detailed requirements further explain specific system behaviors and processes. Each requirement is described clearly to ensure proper interpretation during system design and implementation.

3.3 Intended Audience and Reading Suggestions

This document is intended for developers, project managers, testers, and end users involved in the Airline Flight Data Analysis system. Developers can use this document to understand the system architecture, modules, and design details required for implementation. Project managers can refer to it to understand the project scope, objectives, and overall structure of the system. Testers can use the functional descriptions to create and execute test cases to ensure the system works as expected. End users can read the overview sections to understand the purpose and functionality of the system.

The document is organized into several sections. It begins with an introduction and system overview, followed by the description of system architecture, modules, and design components.

Readers are advised to start with the introduction and overview sections and then continue to the detailed sections based on their role and requirements.

3.4 References The following documents and resources were referred to while preparing this Software Design Specification (SDS) for the Airline Flight Data Analysis system.

1. Python Documentation – Python Software Foundation, Version 3.x, Available at: https://docs.python.org

2. Pandas Library Documentation – Wes McKinney, Pandas Development Team, Available at: https://pandas.pydata.org/docs/

3. NumPy Documentation – NumPy Developers, Available at: https://numpy.org/doc/

4. Matplotlib Documentation – John D. Hunter and Matplotlib Development Team, Available at: https://matplotlib.org/stable/contents.html

5. Software Engineering: A Practitioner's Approach – Roger S. Pressman, McGraw-Hill Education.

6. IEEE Software Requirements Specification (SRS) Standard – IEEE Standard 830 for Software Requirements Specifications.

These references provide guidelines, documentation, and technical information related to software development, data analysis tools, and documentation standards used in the development of this system.

3.5. Analysis Model

3.5. Methodology used

The development of the Airline Flight Data Analysis system follows a function-oriented programming approach using Python. In this approach, the system is divided into different functions, where each function performs a specific task such as data loading, data cleaning, data processing, and visualization. This structure helps in organizing the code efficiently and improves readability and maintainability.

Python libraries such as Pandas, NumPy, and Matplotlib are used for handling datasets, performing data analysis, and generating visual representations of the results. These libraries help in processing large volumes of flight data and extracting meaningful insights from it.

For the development process, the Waterfall Model of the Software Development Life Cycle (SDLC) is used. In this model, the development process is carried out in sequential phases including requirement analysis, system design, implementation, testing, and documentation. Each phase is completed before moving to the next phase, ensuring a structured and systematic development of the software product.

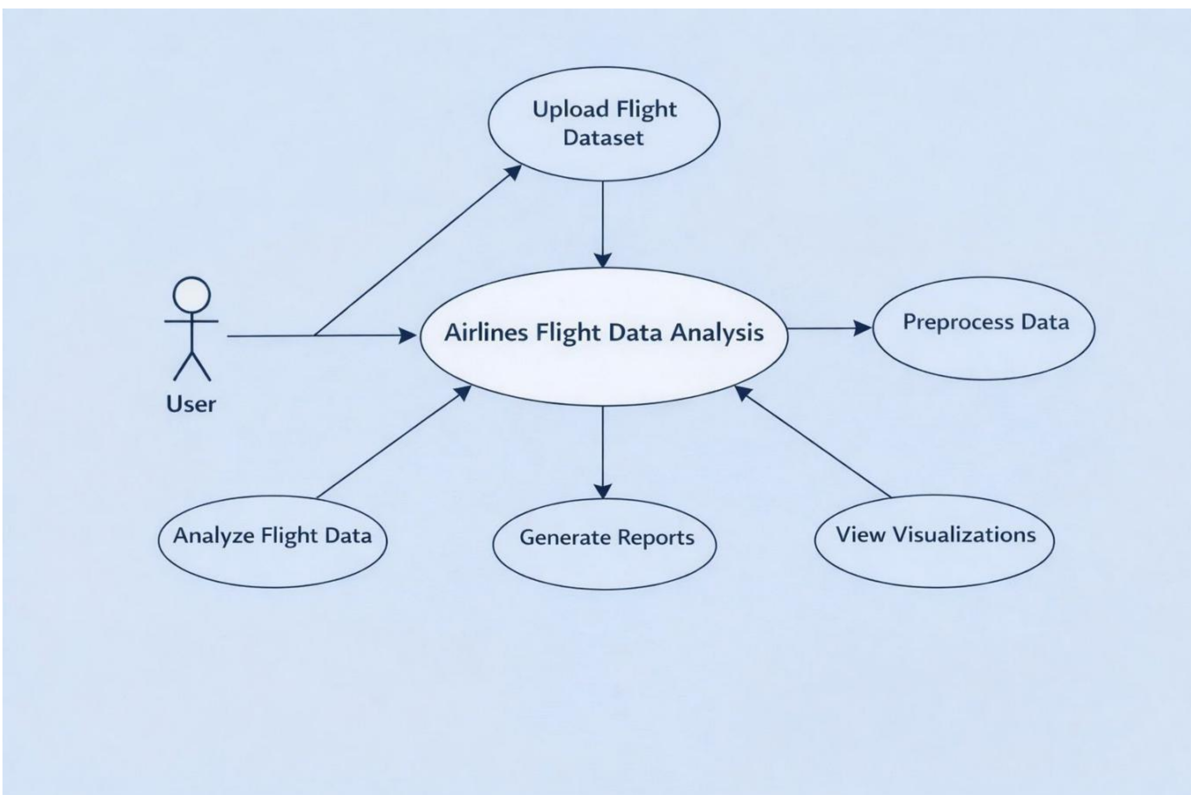
3.5.1 Use Case diagram

Use Case Diagram

A Use Case Diagram is used to represent the interaction between users (actors) and the Airline Flight Data Analysis system. It shows the different actions that users can perform and how the system responds to those actions. The diagram helps in understanding the functional requirements of the system and the relationship between users and system processes.

In this system, the primary actor is the **User/Data Analyst** who interacts with the system to analyze flight data. The main use cases include loading the dataset, cleaning and preprocessing the data, performing data analysis, and generating visualizations or reports based on the analysis.

Each use case represents a specific task performed by the system that provides useful results to the user, such as identifying flight delays, analyzing airline performance, and displaying insights through charts or graphs. The use case diagram helps in clearly illustrating how the user interacts with different functions of the system and how the system processes the data to produce meaningful outputs.



Use Case Specification

Name Of Use Case	Airlines Flight Data Analysis
Actor	User / Data Analyst
Pre Condition	The system is installed and running properly. The flight dataset is available in the system. The user has access to the analysis tools.

Primary Flow of Events	The user starts the system. The user loads the airline flight dataset into the system. The system reads and displays the dataset. The user performs data cleaning and preprocessing. The system processes the data and performs analysis. The system generates charts, graphs, or reports based on the analysis results. The user views and interprets the results.
Alternate Folw Of Events	Alternate Flow 1
	If the dataset is not available or fails to load, the system displays an error message and asks the user to upload the correct dataset.
	Alternate Flow 2
	If the data contains missing or incorrect values, the system prompts the user to clean or preprocess the data before continuing the analysis.
Post Condition	The system successfully analyzes the flight data and displays the results in the form of visualizations or reports.
Use Case termination	The use case ends when the user completes the analysis and closes the system.

3.5.2 ER Model

The Entity-Relationship (ER) Model represents the logical structure of the database used in the Airline Flight Data Analysis system. It shows the main entities, their attributes, and the relationships between them. This model helps in designing a structured database for storing and analyzing flight-related data efficiently.

Entities and Attributes:

1. **Flight**

   Flight_ID (Primary Key)

   Airline_Name

   Flight_Number

   Departure_Airport

   Arrival_Airport

   Scheduled_Departure

   Scheduled_Arrival

Status (On-time/Delayed/Cancelled)

2. **Airline**

Airline_ID (Primary Key)

Airline_Name

Country

Fleet_Size

3. **Airport**

Airport_ID (Primary Key)

Airport_Name

City

Countr

y

4. **Passenger (if tracking bookings or delays affecting passengers)**

Passenger_ID (Primary Key)

Name

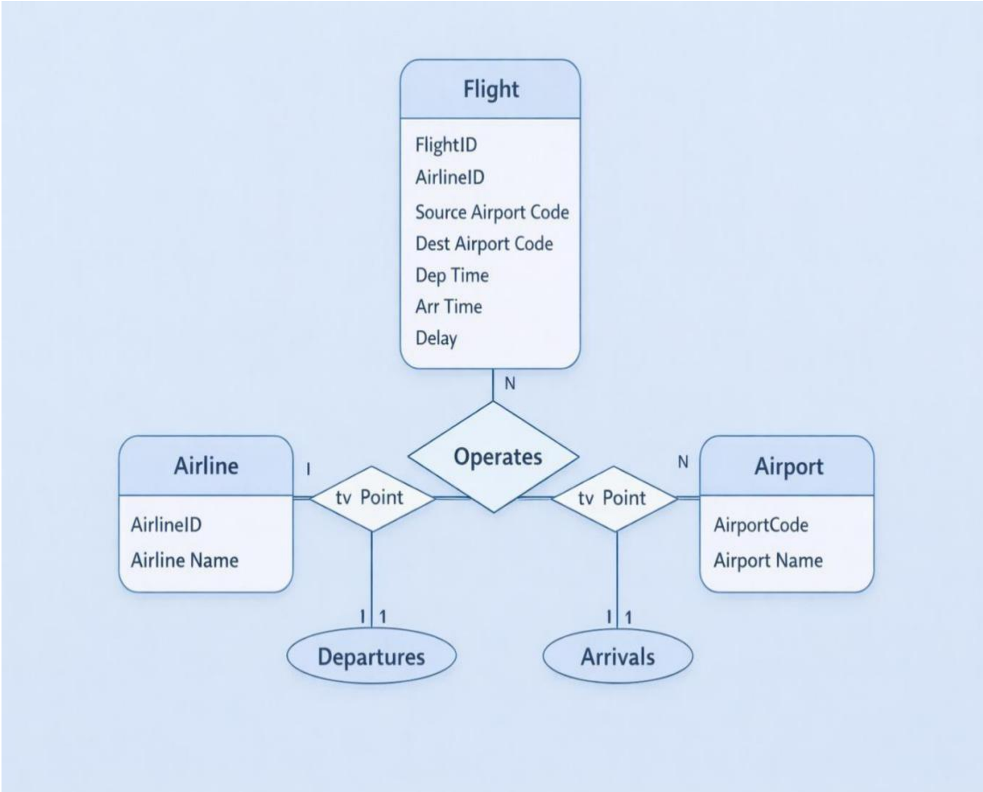
Contact_Info

Flight_ID (Foreign Key)

Relationships:

- Flight – Airline: One airline operates multiple flights (One-to-Many).
- Flight – Airport: Each flight has one departure and one arrival airport (Many-to-One).
- Flight – Passenger: A flight can have multiple passengers (One-to-Many).

This ER model provides a conceptual view of how data is organized and connected in the database, which ensures consistency and supports effective analysis of airline flight data.



3.5.3 Data flow Diagram

A **Data Flow Diagram (DFD)** represents how data moves through a system and how it is processed to produce useful output. It shows the flow of information between different components such as input, processes, data storage, and output. In the **Airlines Flight Data Analysis system**, the DFD explains how flight-related data is collected, processed, analyzed, and finally presented to the user. The system first receives flight dataset information such as airline name, source and destination airports, departure and arrival time, and delay details. This data is then processed and cleaned to remove errors and missing values. After preprocessing, the system performs analysis to identify patterns such as flight delays, airline performance, and popular routes. The processed data may also be stored for future use. Finally, the results are displayed in the form of reports, charts, or graphs to help users understand the insights obtained from the flight data analysis.

Process Specification of Significant Processes

1. Data Collection:
This process involves gathering flight data from available datasets. The data contains information about airlines, flight schedules, routes, and delay status.

2. Data Pre-processing:
In this stage, the collected data is cleaned and organized. Missing values are handled and unnecessary or incorrect data is removed to make the dataset suitable for analysis.

3. Data Analysis:
The system analyzes the cleaned dataset to find useful patterns such as delay trends, busiest airports, and airline performance.

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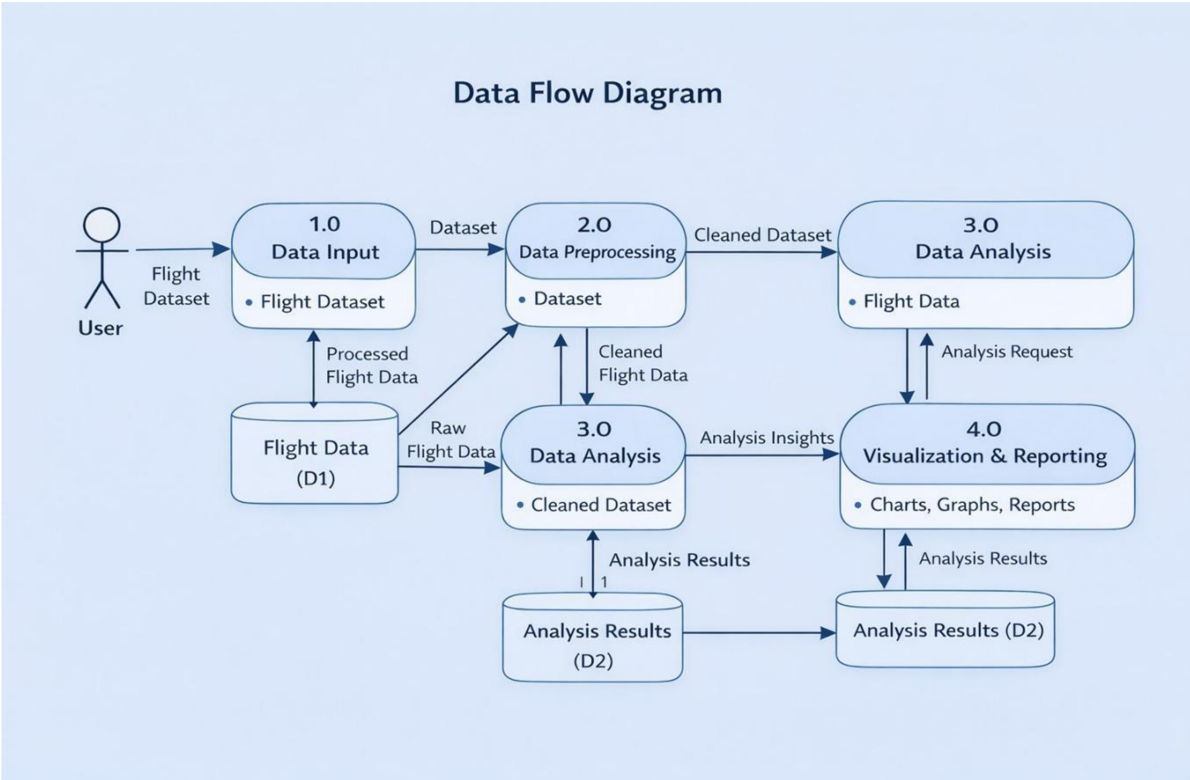
6. Data Analysis:
The system analyzes the cleaned dataset to find useful patterns such as delay trends, busiest airports, and airline performance.

7. Result Visualization:
The final analyzed data is presented using charts, graphs, and summary reports so that users can easily understand the results of the flight data analysis.



3.5.5 Control Flow Diagram

A **Control Flow Diagram (CFD)** represents the flow of control within a system and shows how different processes are executed step by step. It focuses on the sequence of operations and decision points that control the execution of tasks. In the **Airlines Flight Data Analysis system**, the control flow begins when the user provides the flight dataset for analysis. The system then checks the data and moves to the preprocessing stage where the dataset is cleaned and prepared. After preprocessing, the control is passed to the data analysis stage where different analytical operations are performed to identify patterns such as delays, airline performance, and route statistics. Based on the processed data, the system generates visualizations and reports. The control flow ensures that each process is executed in the correct order so that the final output is accurate and meaningful.

Control Specification of Significant Processes

1. Data Input Control:
The system first checks whether the dataset is available and correctly loaded. If the dataset is missing or invalid, the system stops the process and asks for valid input.

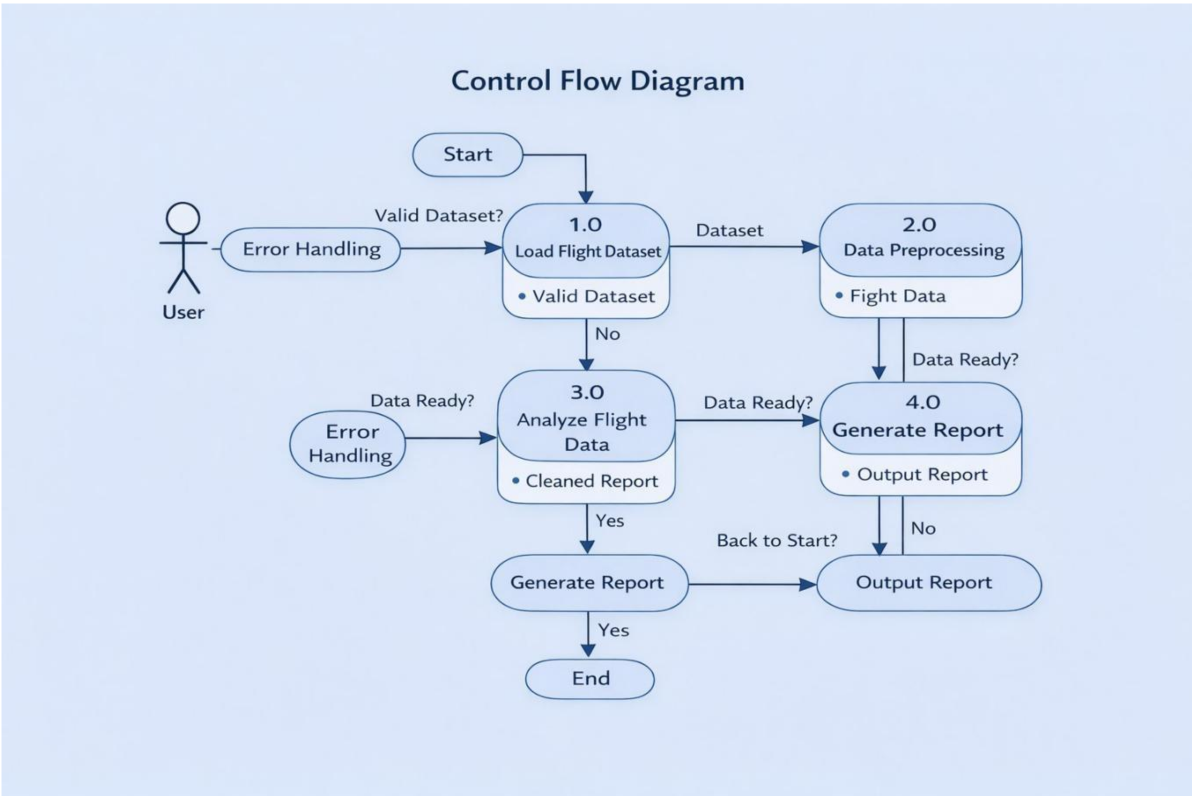
2. Preprocessing Control:
Once the dataset is loaded, the system performs data cleaning and validation. Control moves to the next step only when the dataset is properly prepared.

3. Analysis Control:

The system performs various analysis operations on the processed data. This step identifies trends such as delay frequency, airline performance, and route statistics.

4. Output Generation Control:

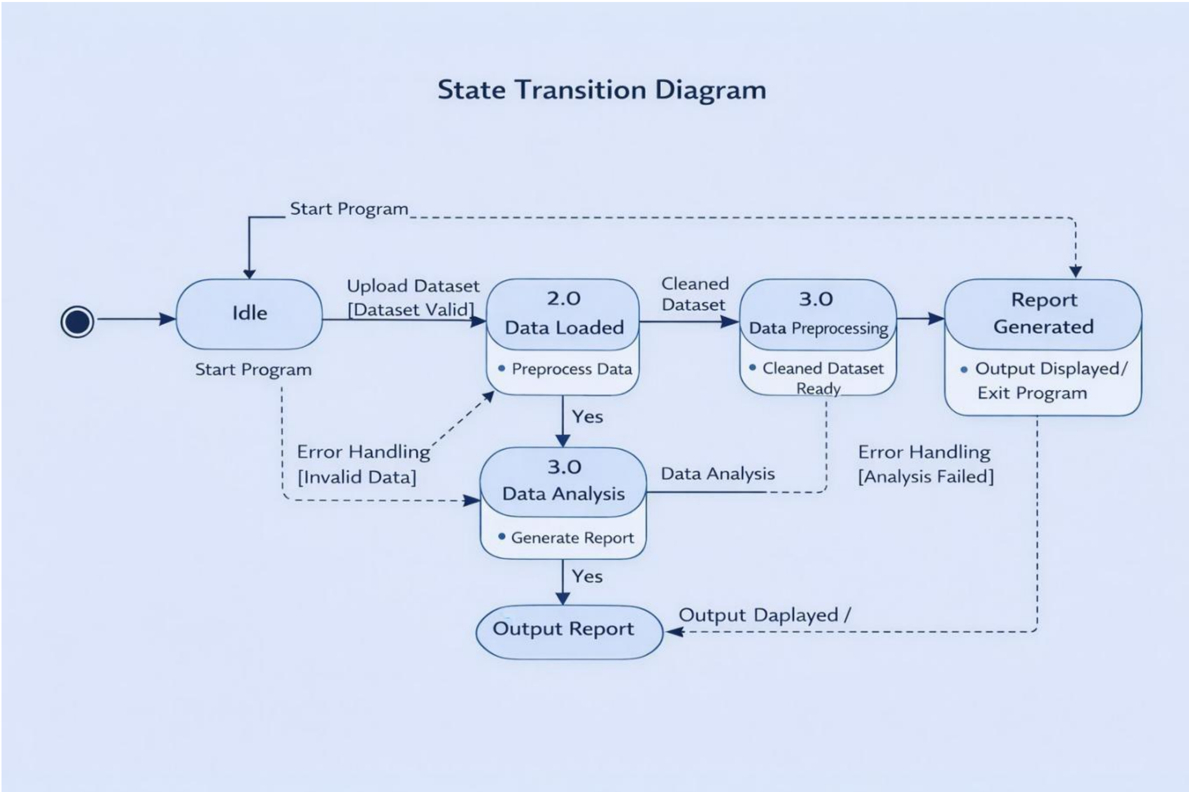
After successful analysis, the system generates results in the form of charts, graphs, or reports. These outputs help users understand the insights obtained from the flight data.



3.5.6 State Transition Diagram

A **State Transition Diagram** describes the different states of an object in a system and how the system changes from one state to another based on events and conditions. It helps in understanding the behavior of the system during different stages of operation. In the **Airlines Flight Data Analysis system**, the process begins with the **initial state**, where the system waits for the user to provide the flight dataset. Once the dataset is uploaded, the system moves to the **data loading state**, where the data is read and prepared for processing. After the data is successfully loaded, the system enters the **data preprocessing state**, where missing values and errors in the dataset are handled to make the data suitable for analysis.

After preprocessing, the system transitions to the **data analysis state**, where different analytical operations are performed to identify useful information such as flight delay patterns, airline performance, and route statistics. When the analysis is completed, the system moves to the **result generation state**, where the results are displayed in the form of charts, graphs, or reports. Finally, the system reaches the **final state**, indicating that the flight data analysis process has been successfully completed.



State Transition Diagram
Start Program
Idle
Upload Dataset
[Dataset Valid]
2.0
Data Loaded
Preprocess Data
Cleaned
Dataset
3.0
Data Preprocessing
Cleaned Dataset
Ready
Report
Generated
Output Displayed/
Exit Program
Start Program
Yes
Error Handling
[Invalid Data]
3.0
Data Analysis
Generate Report
Data Analysis
Error Handling
[Analysis Failed]
Yes
Output Report
Output Daplayed /

CHAPTER 4: IMPLEMENTATION / PRESENT INVESTIGATION

4.1 Introduction

This chapter presents the detailed implementation of the flight data analysis project. The dataset obtained from Kaggle is processed using Python libraries, and various analytical techniques are applied to extract meaningful insights.

The implementation phase includes data loading, data cleaning, feature engineering, and visualization. The primary objective of this chapter is to analyze different factors affecting flight ticket prices such as airline, departure time, arrival time, and booking time.

4.2 Data Loading and Exploration

The dataset was loaded into the Python environment using the Pandas library. After loading, the structure of the dataset was examined to understand the available features and identify missing values.

✔ Steps Performed:

- Importing dataset
- Checking dataset structure
- Identifying null values
- Understanding column types

```
import pandas as pd
df = pd.read_csv("flights.csv")
df.head()
df.info()
```

The dataset consists of multiple attributes such as airline name, source, destination, departure time, arrival time, duration, number of stops, and ticket price.

4.3 Data Preprocessing

Before performing analysis, the dataset was cleaned and preprocessed to ensure accuracy and consistency.

✔ Preprocessing Steps:

4.3.1 Handling Missing Values

Missing values were identified and handled appropriately. Rows with excessive missing data were removed.

4.3.2 Removing Duplicates

Duplicate records were removed to avoid redundancy and ensure correct analysis.

4.3.3 Data Type Conversion

Columns such as date and time were converted into appropriate formats.

4.3.4 Feature Engineering

New features were derived from existing data. For example:

- Extracting day and month from date
- Categorizing time into slots (Morning, Evening, etc.)

These steps improved the quality of the dataset and made it suitable for analysis

4.4 Analysis Based on Airline

To understand how ticket prices vary across different airlines, a comparative analysis was performed.

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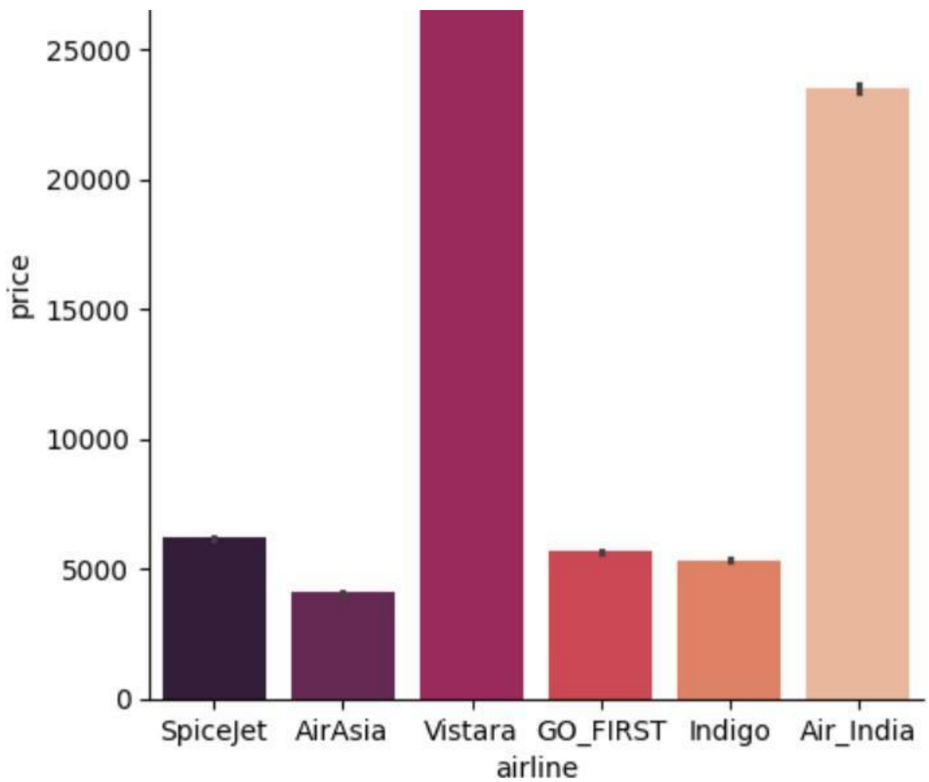

Figure 4.1: Airline vs Ticket Price

Explanation:

The above graph shows the variation in ticket prices among different airlines. It can be observed that certain airlines have higher ticket prices compared to others. This variation may be due to factors such as airline reputation, quality of service, availability of direct flights, and operational efficiency.

Premium airlines generally charge higher prices, while low-cost carriers offer more affordable options. This analysis helps in understanding pricing strategies adopted by airlines.

4.5 Analysis Based on Departure Time

To analyze the impact of departure time on ticket prices, the dataset was grouped based different

Time slots.

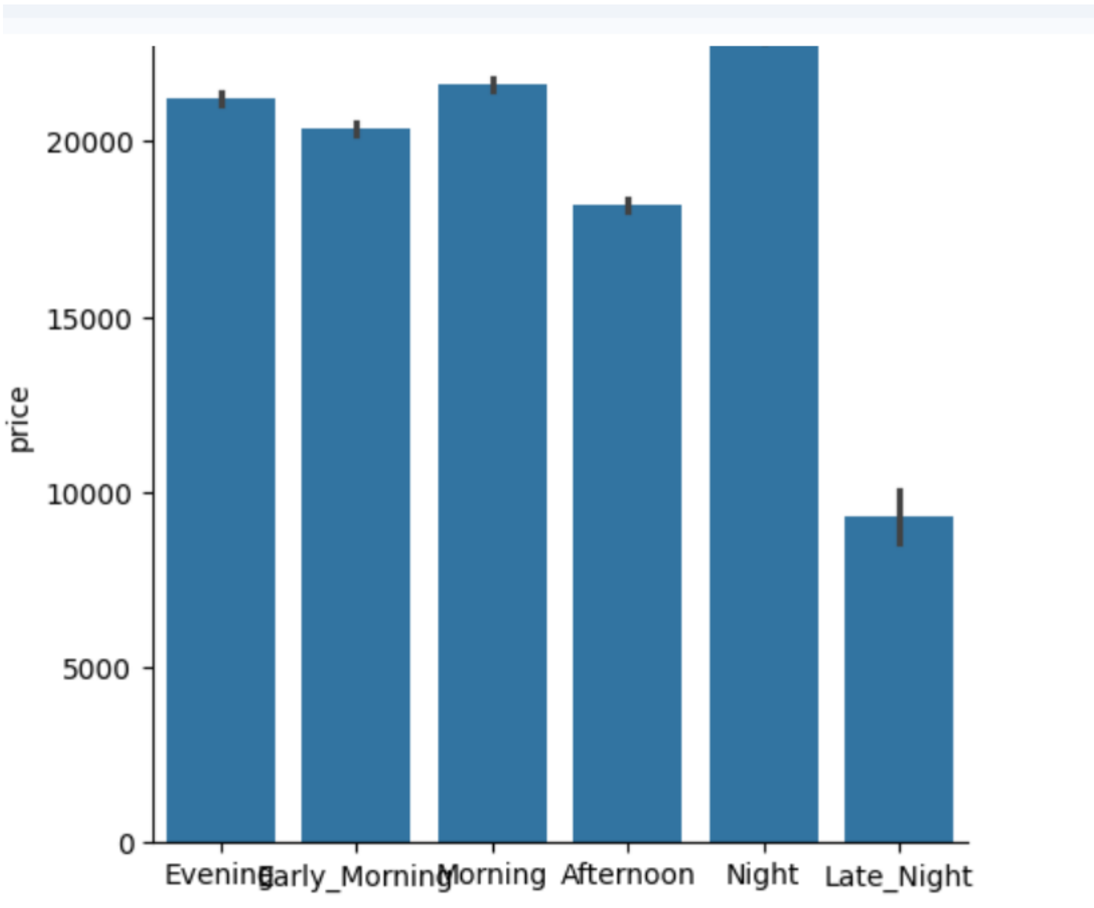

Figure 4.2: Departure Time vs Ticket Price

Explanation:

The graph illustrates how ticket prices vary across different departure times such as early morning, morning, afternoon, evening, night, and late night.

It is observed that flights during peak hours (morning and evening) tend to have higher prices due to increased demand. Late-night flights are relatively cheaper as fewer passengers prefer traveling during those hours.

4.6 Analysis Based on Arrival Time

Arrival time also influences ticket pricing as it affects passenger convenience.

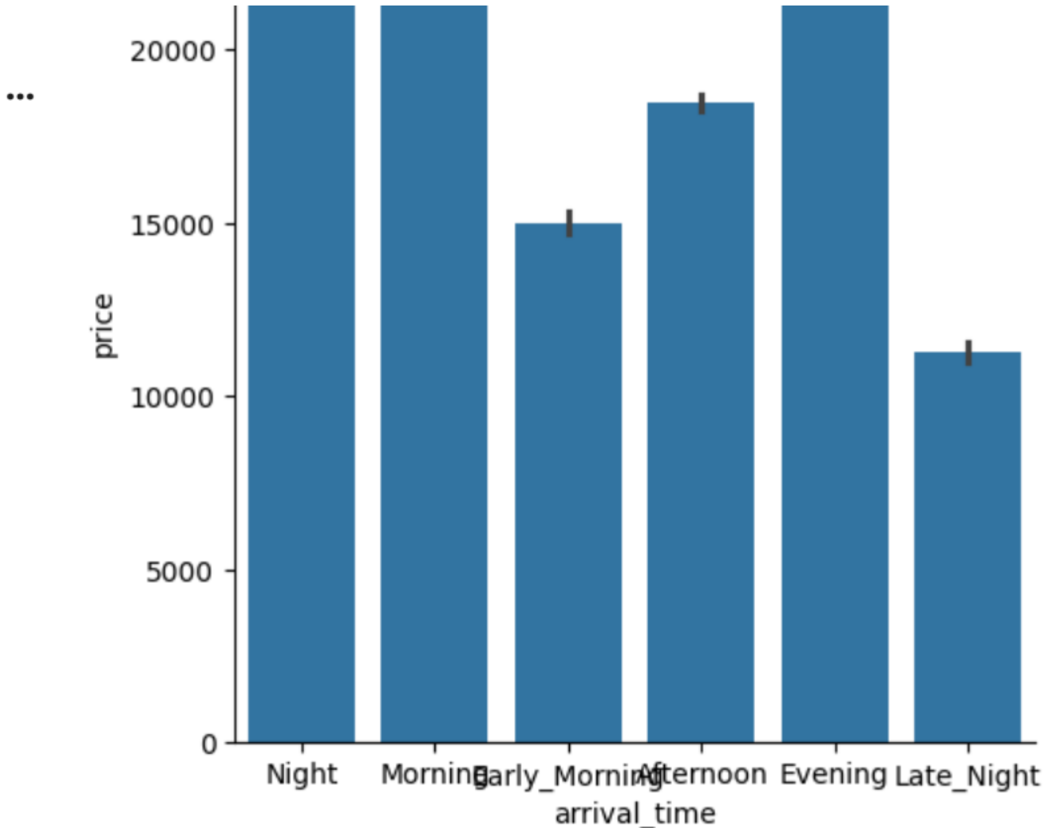

Figure 4.3: Arrival Time vs Ticket Price

Explanation:

The graph shows the variation in ticket prices based on arrival time. Flights arriving during peak hours such as morning and evening have higher ticket prices. On the other hand, flights arriving late at night are cheaper.

This indicates that airlines adjust their pricing based on passenger preferences and convenience.

4.7 Combined Analysis of Departure and Arrival Time

A combined analysis was performed to understand the relationship between departure time, arrival time, and ticket price.

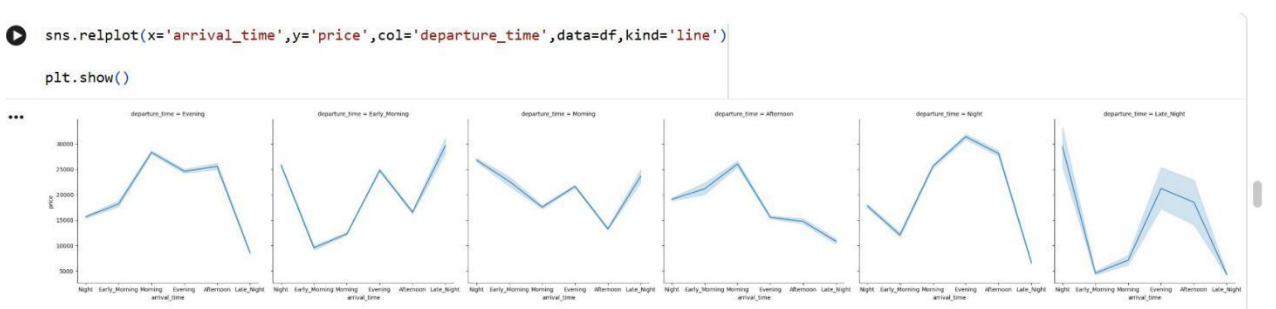

Figure 4.4: Combined Analysis of Departure and Arrival Time

Explanation:

This figure represents multiple comparisons of departure and arrival times with respect to ticket price. It provides a deeper understanding of how different combinations of travel timings affect pricing.

Certain combinations, such as morning departure with evening arrival, tend to have higher prices due to convenience and demand. Less preferred combinations, such as late-night travel, are more affordable.

4.9 Analysis Based on Source and Destination City

To understand how ticket prices vary across different cities, an analysis was performed based on source and destination locations. This helps in identifying high-demand routes and expensive travel paths.

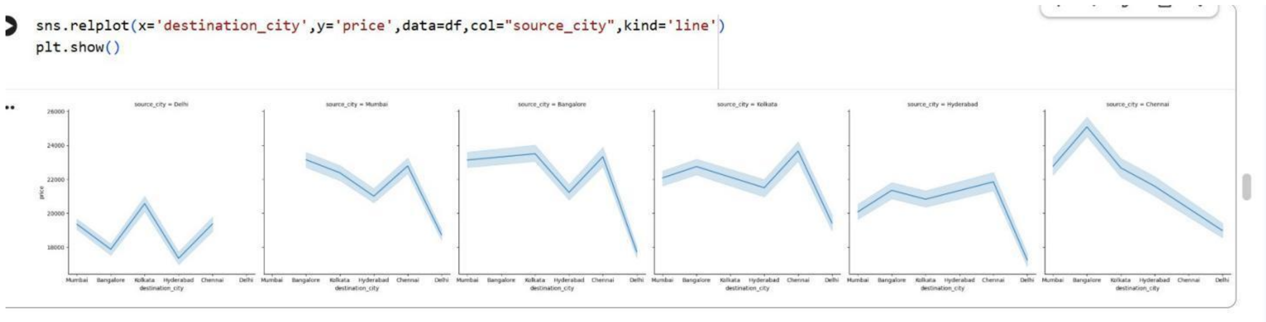

Figure 4.6: Source and Destination City vs Ticket Price

Detailed Explanation:

The above graph represents the variation in ticket prices based on different source and destination cities. It can be observed that certain routes have significantly higher prices compared to others.

This variation is mainly due to factors such as:

- Distance between cities
- Demand for specific routes
- Availability of flights
- Popular travel destinations

Routes connecting major metropolitan cities tend to have higher ticket prices due to high demand. On the other hand, less popular routes are relatively cheaper.

This analysis helps in understanding how geographical factors influence airline pricing strategies.

4.10 Analysis Based on Days Left Before Departure

To understand the effect of booking time on ticket prices, the number of days left before departure was analyzed.

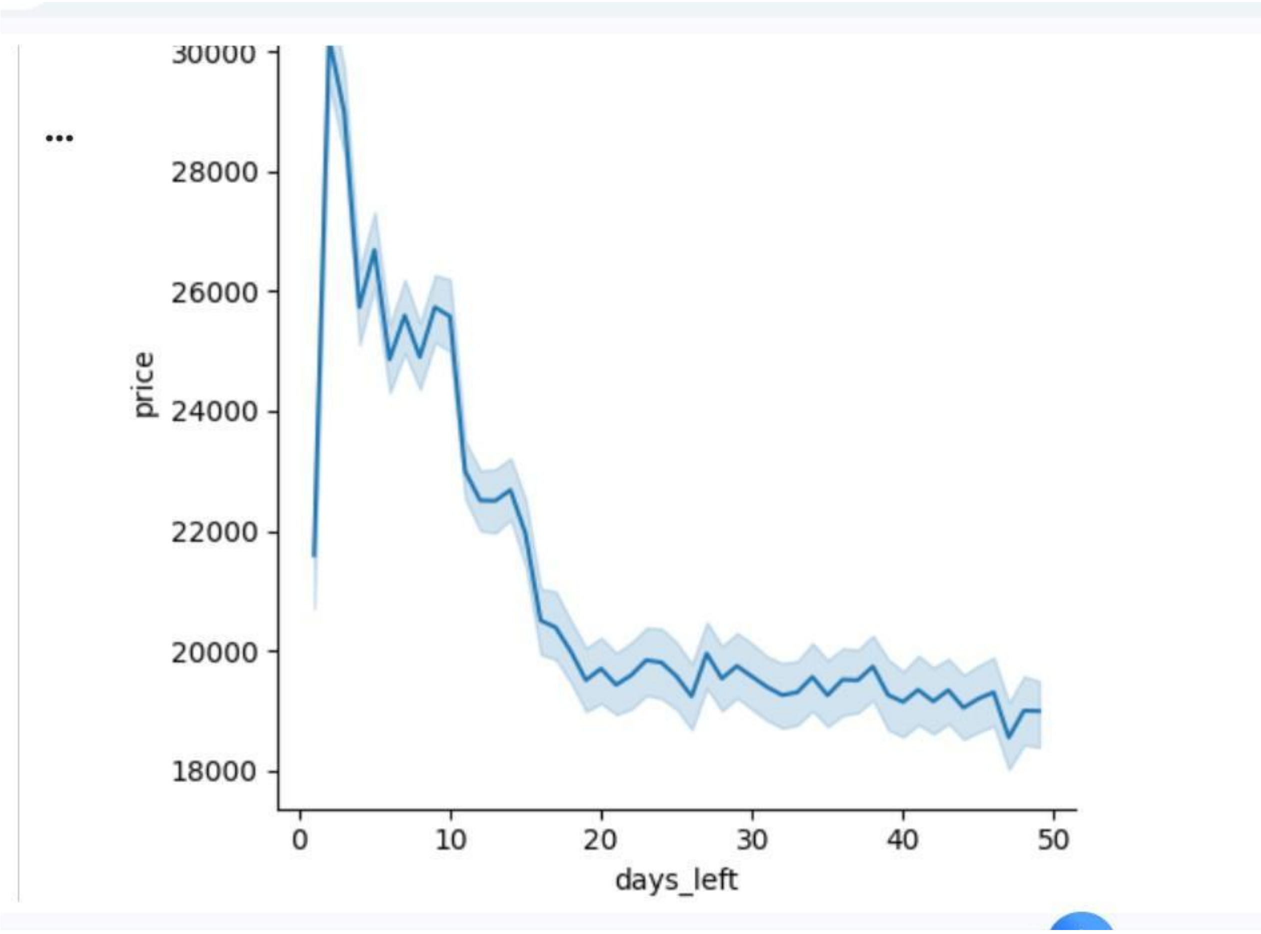

Figure 4.5: Days Left vs Ticket Price

Explanation:

The graph shows the relationship between the number of days left before departure and ticket price. It is clearly observed that ticket prices increase as the departure date approaches.

Passengers who book tickets earlier get lower prices, while last-minute bookings are more expensive. This trend reflects demand-based pricing strategies used by airlines.

4.11 Summary of Implementation

In this chapter, the dataset was successfully processed and analyzed using Python. Various factors affecting flight ticket prices were studied using graphical representations.

The results obtained from these analyses provide a clear understanding of pricing trends in the aviation industry. These insights will be further discussed in the next chapter.

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents the results obtained from the flight data analysis performed in the previous chapter. The findings are discussed in detail to highlight patterns, trends, and relationships between different variables such as airline, departure time, arrival time, number of stops, booking time, and route.

The objective of this chapter is to interpret the results obtained from graphical analysis and draw meaningful conclusions from the dataset.

5.2 Analysis of Airline Pricing

From **Figure 4.1**, it is observed that different airlines follow different pricing strategies. Some airlines consistently offer higher ticket prices, while others provide more economical options.

Premium airlines tend to charge higher prices due to better services, fewer stops, and higher reliability. Budget airlines, on the other hand, focus on affordability and attract price-sensitive customers.

This variation clearly indicates that airline brand and service quality play an important role in ticket pricing.

5.3 Impact of Departure Time

Based on **Figure 4.2**, ticket prices vary significantly with departure time. Flights scheduled during peak hours such as morning and evening are more expensive.

This is mainly due to higher demand during these time periods, as most passengers prefer convenient travel times. Late-night flights are comparatively cheaper because fewer passengers prefer traveling during those hours.

Thus, departure time is a key factor influencing ticket prices.

5.4 Impact of Arrival Time

From **Figure 4.3**, it is evident that arrival time also affects ticket pricing. Flights arriving during peak hours tend to have higher prices.

Passengers prefer flights that arrive at convenient times, such as early morning or evening, which leads to increased demand and higher prices. Flights arriving late at night are less expensive due to lower demand.

5.5 Combined Effect of Departure and Arrival Time

From **Figure 4.4**, it can be concluded that the combination of departure and arrival times has a significant impact on ticket prices.

Certain combinations, such as morning departure and evening arrival, are priced higher due to convenience and high demand. Less preferred combinations, such as late-night travel, are relatively cheaper.

This shows that airlines optimize pricing based on both departure and arrival timings.

5.6 Effect of Booking Time (Days Left)

From **Figure 4.5**, a clear trend can be observed between ticket price and the number of days left before departure.

Ticket prices are lower when booked well in advance and increase gradually as the departure date approaches. This is due to demand-based pricing strategies and limited seat availability.

This analysis suggests that early booking is beneficial for passengers seeking lower prices.

5.7 Impact of Source and Destination Cities

From **Figure 4.6 / 4.7**, it is observed that ticket prices vary significantly across different routes.

Routes connecting major cities or popular destinations tend to have higher prices due to high demand. In contrast, less popular routes are relatively cheaper.

Distance between cities and availability of flights also influence ticket pricing.

5.8 Key Findings

The major findings of this project are:

- Ticket prices vary significantly across airlines
- Peak hour flights are more expensive
- Late-night flights are cheaper
- Prices increase as departure date approaches
- Popular routes have higher ticket prices
- Combination of departure and arrival time affects pricing

5.9 Discussion

The results obtained from this project clearly indicate that flight ticket pricing is influenced by multiple factors. Airlines use dynamic pricing strategies based on demand, timing, and route popularity.

Passengers who prioritize convenience are willing to pay higher prices, while cost-sensitive travelers opt for cheaper alternatives such as late-night flights or flights with stops.

The use of data visualization techniques made it easier to identify these patterns and trends.

5.10 Limitations of the Study

- The dataset used is limited to historical data
- Real-time factors such as weather were not considered
- No predictive models were implemented
- Analysis is limited to available features in the dataset

5.11 Future Scope

- Implementation of machine learning models for price prediction
- Use of real-time flight data
- Development of an interactive dashboard
- Inclusion of additional factors such as weather and holidays

CHAPTER 6: SUMMARY AND CONCLUSION

6.1 Summary of the Work

This project focused on analyzing flight data using Python in order to extract meaningful insights related to airline operations and ticket pricing. The dataset used in this project was obtained from Kaggle and contained various attributes such as airline name, source and destination cities, departure time, arrival time, duration, number of stops, and ticket price.

The project began with data collection and preprocessing, where the dataset was cleaned by handling missing values, removing duplicate records, and converting data into appropriate formats. Feature engineering techniques were also applied to derive new attributes such as time categories and date-related features.

After preprocessing, exploratory data analysis was performed using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn. Various graphs and visualizations were created to understand patterns and relationships in the data.

The analysis focused on multiple factors affecting ticket prices, including airline, departure time, arrival time, number of stops, booking time, and route. The results obtained from these analyses provided valuable insights into pricing strategies used by airlines.

6.2 Conclusion

From the analysis performed in this project, several important conclusions can be drawn:

- Ticket prices vary significantly across different airlines due to differences in service quality and brand value.
- Flights during peak hours such as morning and evening are more expensive due to higher demand.
- Late-night flights are generally cheaper as fewer passengers prefer traveling during those hours.
- Ticket prices increase as the departure date approaches, highlighting the importance of early booking.
- Routes connecting major cities tend to have higher prices due to increased demand.
- The combination of departure and arrival time also plays a significant role in determining ticket prices.

Overall, the project successfully demonstrated how data analysis techniques can be used to extract meaningful insights from large datasets. The use of visualization tools made it easier to identify patterns and trends in the data.

6.3 Contributions of the Project

The main contributions of this project are:

- Performed detailed exploratory data analysis on flight data
- Identified key factors affecting ticket pricing
- Used visualization techniques to present insights clearly
- Provided a structured approach for analyzing large datasets

This project can serve as a foundation for further research in the field of flight data analysis.

6.4 Limitations of the Study

Despite the successful analysis, the project has some limitations:

- The dataset used is limited to historical data
- Real-time factors such as weather conditions were not considered
- No predictive models were implemented
- Analysis is limited to available features in the dataset

These limitations can be addressed in future work.

6.5 Future Scope

The project can be further extended in several ways:

- Implementation of machine learning models for ticket price prediction
- Use of real-time flight data for more accurate analysis
- Development of an interactive dashboard for visualization
- Inclusion of additional factors such as weather conditions and holidays
- Integration with web applications for real-time user interaction

6.6 Final Remarks

This project highlights the importance of data analysis in the aviation industry. By analyzing flight data, valuable insights can be obtained that help in improving airline operations and decision-making.

The techniques used in this project can be applied to other domains as well, making it a versatile and practical approach to data analysis.